

Topic 7: Audio, Video, and Multimodal AI

7.1 AI Audio Generation

- Text-to-speech (TTS): AI generates natural-sounding speech from text -- far more natural than older synthesisers
- ElevenLabs: high-quality TTS and voice cloning; clone a voice from a short audio sample (elevenlabs.io)
- OpenAI TTS: available via API; voices include Alloy, Echo, Fable, Onyx, Nova, Shimmer
- Whisper (OpenAI): state-of-the-art speech-to-text; open source and free to use (github.com/openai/whisper)
- AI music generation: Suno (suno.com) and Udio (udio.com) generate full songs with lyrics from a text prompt
- Audio tools are rapidly improving -- voice cloning in particular raises serious concerns about deepfake audio

7.2 AI Video Generation

- AI video generation produces short clips from text prompts or images -- quality improves rapidly
- Sora (OpenAI): generates realistic, temporally consistent videos up to one minute; available to ChatGPT Pro users
- Runway Gen-3 Alpha: professional-grade AI video generation and editing tools (runwayml.com)
- Kling AI (Kuaishou): high-quality video generation; free tier available (klingai.com)
- Pika: generates and edits short video clips from text or image prompts (pika.art)
- Limitation: AI video still struggles with physics, consistent hands/faces, and long coherent narratives

7.3 Multimodal AI Systems

- Multimodal AI: models that accept and generate multiple types of data (text, image, audio, video) in one system
- GPT-4o (OpenAI): accepts text, image, and audio input; generates text and audio; powers ChatGPT voice mode
- Claude 3.5 (Anthropic): accepts text and images; particularly strong at analysing documents and charts
- Gemini 1.5 Pro (Google): natively multimodal; can process text, images, audio, video, and documents
- Use cases: describe a chart image and get analysis, take a photo of code and ask for a review, speak and receive spoken reply
- Multimodal models are converging text, vision, and audio into single unified systems

7.4 Deepfakes and Synthetic Media Risks

- Deepfakes: AI-generated or AI-manipulated video/audio of real people -- voice cloning and face-swapping are now accessible
 - Risks: non-consensual content, political disinformation, financial fraud via fake CEO audio/video
 - Detection tools exist but are not fully reliable: Microsoft's Video Authenticator, Hive Moderation
 - Legal landscape: several countries and US states now have laws specifically targeting non-consensual deepfakes
 - Best practice: verify surprising audio or video of public figures against multiple reliable sources before sharing
 - Content credentials (C2PA standard): a technical framework for signing media with proof of human or AI origin
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